B13 Series B

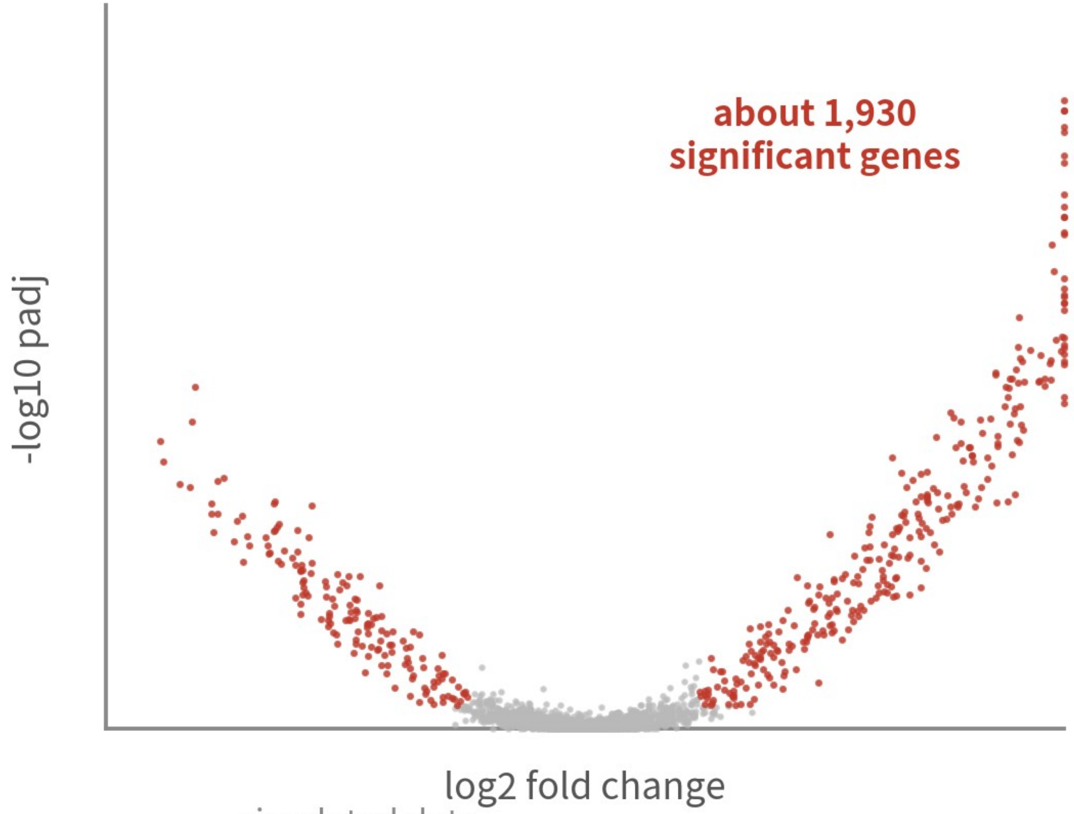
After DE: functional enrichment and gene set analysis

Turn 1,930 genes into one biological conclusion — without falling into the classic enrichment traps.

About 40 min | Prerequisites: B12 | Data: ifnb (B12 DE results)

After the volcano: the reviewer's question

the B12 pseudobulk volcano (CD14 Mono)



simulated data

IFIT1 IFI6 LY6E
IFI35 STAT1 LY6E
BST2 RSAD2 IFITM3
BST2 LY6E IFI6
SAMD9 IFITM3 CXCL10
STAT1 BST2 IFITM3
PLS
BST
IFI
...

Reviewer:
"So what actually happened, biologically?"

reading out 1,930 names is not an answer

B12 handed you 1,930 significant genes. Now what?

WHERE THIS EPISODE STARTS

A gene list is not a conclusion. Between the list and the biology, there is one more step.

That step is enrichment: asking what your list looks like, against decades of curated gene sets.

What you will learn

1. Tell ORA and GSEA apart — what each asks, and when to use which
2. Run fgsea + Hallmark on the B12 DE results and read NES and leading edges
3. Score cells with AddModuleScore / AUCell and interpret the scores correctly

01

Gene sets: prior knowledge, packaged

Two ways to ask: ORA asks the list, GSEA asks the ranking

Four databases you will keep meeting

GO	KEGG	Reactome	MSigDB
a controlled vocabulary three branches: BP / MF / CC	hand-drawn pathway maps metabolism and signalling	reaction-level pathways expert-curated, open	the gene-set supermarket Hallmark is its curated shelf
GO Consortium tens of thousands of terms	KEGG (Kyoto) hundreds of pathways	Reactome team signalling to metabolism	Broad / UCSD 30k+ sets in collections

a gene set = a list of genes sharing a functional label — decades of biology, packaged

They all package the same thing: a gene list with a functional label

MSigDB and Hallmark: the opener

MSigDB collections (selected)

Collection	Description
H	Hallmark: 50 curated sets
C2	curated: KEGG, Reactome···
C5	ontology: the GO terms
C7	immunologic signatures
C8	cell type signatures

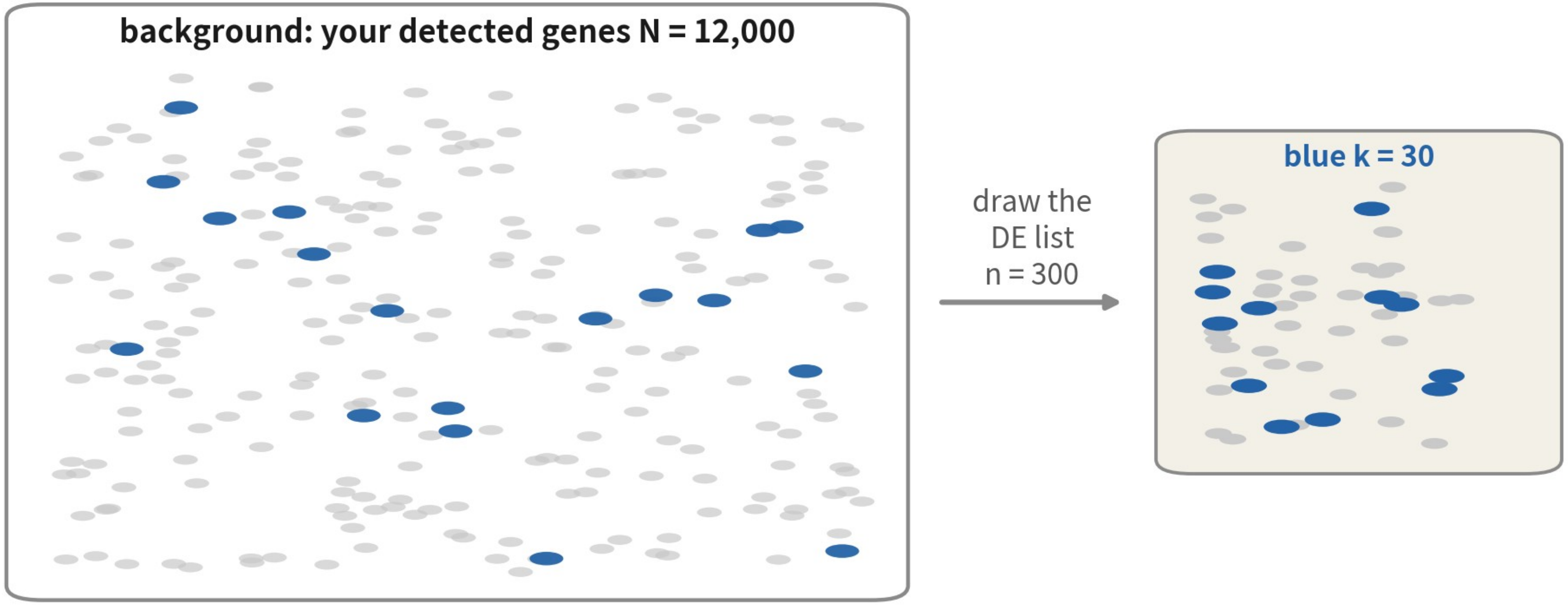
why Hallmark is the default opener

- ✓ just 50 sets — one readable table
- ✓ deliberately de-duplicated: one process, one set
- ✓ plain names: INTERFERON_ALPHA_RESPONSE
- ✓ set direction first; dig details in GO / Reactome

e.g. HALLMARK_INTERFERON_ALPHA_RESPONSE (97 genes)

- MSigDB collects gene sets into one place, organized in collections
- Hallmark (collection H): 50 curated, deliberately de-duplicated sets
- Set direction with Hallmark first; dig details in GO / Reactome
- R access: the msigdbr package, one call away

ORA: the urn model



blue = pathway members K = 85

the hypergeometric test asks:

drawing 300 at random expects ≈ 2.1 blue; you got 30

probability of this $p \approx 5.0e\text{-}27$

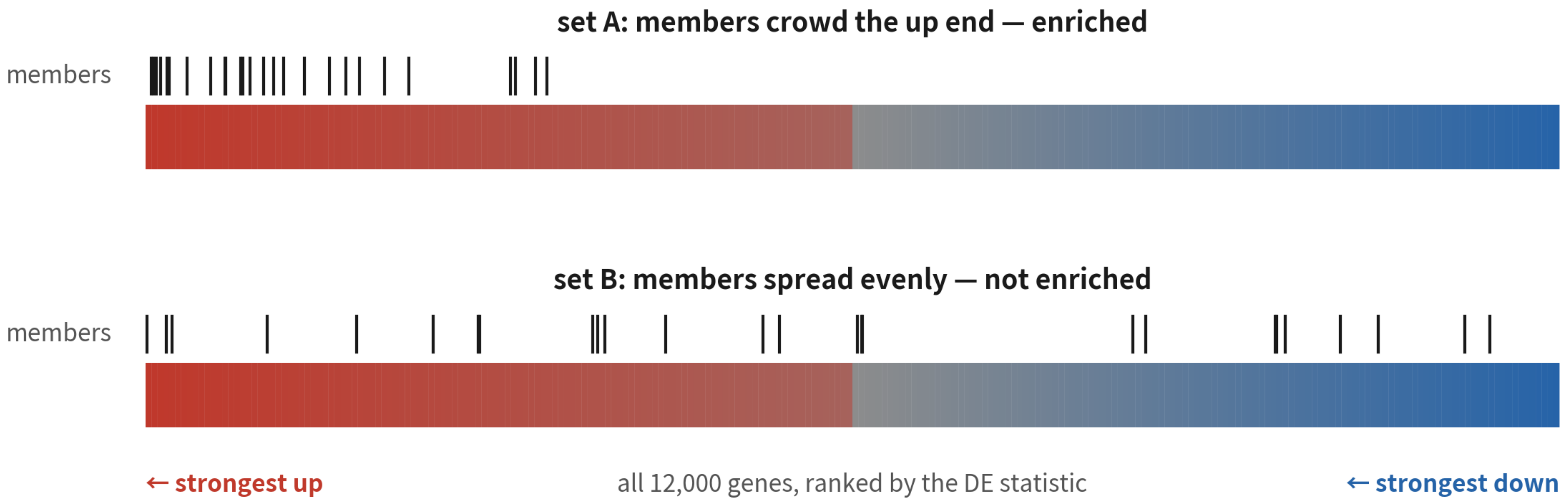
p computed for real

ORA eats only the list — who is on it, and what the background is, decides p

The hypergeometric test: does my list hold more members than random drawing would?

GSEA: no cutoff — use the whole ranking

simulated data



GSEA needs no cutoff — it asks whether members pile up at one end of the ranking

It asks whether a set's members collectively crowd one end of the ranking

ORA vs GSEA: when to use which

Not a religion — your input decides

ORA

You only have a list

- input: a thresholded list + a background
- hypergeometric / Fisher test
- cutoff-sensitive; a wrong background poisons everything

GSEA

You have the full ranking

- input: a ranking statistic for all genes
- no cutoff needed — weak coordinated signals count
- this episode's mainline: a DE table is born ranked

CORE IDEA

Enrichment never tests your data — it tests your list against a background.

How the list was made and what the background is — these decide what the result means. All three traps grow from here.

02

From DE results to a ranked list

Load the B12 output, build one ranking vector

Load the B12 DE results

```
library(Seurat)
library(dplyr)
set.seed(1234)            # 全系列固定，結果可重現

de <- readRDS("output/ifnb_b12_de_results.rds")
names(de)
pb <- de$pseudobulk         # DESeq2 shrink 後的結果表
head(pb[order(pb$padj), ], 3)
```

```
[1] "naive"    "pseudobulk" "coldata"   "audit"
   baseMean log2FoldChange lfcSE  pvalue   padj
ISG15  1520.3       5.08 0.211 2.1e-31 3.4e-27
IFIT1   980.6       5.42 0.243 8.9e-30 7.2e-26
IFIT3  1213.9       4.61 0.222 5.0e-29 2.7e-25
```

Of these columns, today we need log2FoldChange and pvalue

The ranking metric: both camps, spelled out

the B12 DE table

gene	log2FC	pvalue
ISG15	5.1	2e-31
IFIT3	4.6	8e-29
CXCL10	4.9	3e-24
SELL	-1.9	6e-09
RPL9	-0.1	0.62

one score per gene →

ranking metric

`sign(log2FC) × (-log10 p)`

up & significant → large +
down & significant → large –

sort descending →

the ranked list (named vector)

ISG15	+30.7
IFIT3	+28.1
CXCL10	+23.5
…	
RPL9	+0.21
…	
SELL	-8.2

the other camp ranks by the DESeq2 Wald stat — nearly identical; report shrunken FC, rank by the statistic

Direction times significance, or the Wald stat directly. Rank by the statistic; report shrunken FC

Build the ranks

```r
pb <- pb[!is.na(pb$pvalue) & !is.na(pb$log2FoldChange), ]
# 方向 × 顯著程度; p = 0 的基因以 1e-300 保底再取 log
pb$metric <- sign(pb$log2FoldChange) *
        -log10(pmax(pb$pvalue, 1e-300))
ranks <- sort(setNames(pb$metric, rownames(pb)),
        decreasing = TRUE)
length(ranks); head(ranks, 3); tail(ranks, 2)
```

```
[1] 11143
ISG15  IFIT1  IFIT3
 30.7   29.1   28.3
 CCL2   SELL
-10.8  -12.4
```

One named vector: gene names, ranking scores, sorted descending

Three checks before fgsea

Length

11,143 genes — everything tested, not just the significant

Who sits at the ends

IFN genes on top, down-regulated at the tail — direction is right

No NA, no duplicate names

fgsea tolerates neither — clean first

03

Running fgsea with Hallmark

Known-answer testing: on IFN-beta data, what must light up?

Fetch Hallmark with msigdbr

```r
# install.packages("msigdbr")
library(msigdbr)
hm <- msigdbr(species = "Homo sapiens", collection = "H")
# 舊版 msigdbr 的參數名是 category = "H"，內容相同
hallmark <- split(hm$gene_symbol, hm$gs_name)
length(hallmark)
sapply(hallmark[1:2], length)
```

```
[1] 50
   HALLMARK_ADIPOGENESIS HALLMARK_ALLOGRAFT_REJECTION
                     200                          207
```

After split it is exactly what fgsea wants: a list of gene-name vectors

Run fgsea

```r
# BiocManager::install("fgsea")
library(fgsea)
set.seed(1234)

gsea <- fgsea(pathways = hallmark,
              stats   = ranks,
              minSize = 15,  # 太小的集：估不穩
              maxSize = 500, # 太大的集：什麼都沾一點
              eps     = 0)   # p 值算到底，不設下限
```

All three have defaults; these are common conventions — reasons in the narration

The results: check the top first

```
gsea %>% arrange(padj) %>%
 select(pathway,NES,padj,size) %>% head(5)
```

```
                pathway NES   padj size
1 HALLMARK_INTERFERON_ALPHA_RESPONSE 2.88 4.1e-23  95
2 HALLMARK_INTERFERON_GAMMA_RESPONSE 2.79 4.1e-23 186
3   HALLMARK_INFLAMMATORY_RESPONSE 1.94 1.2e-05 169
4   HALLMARK_IL6_JAK_STAT3_SIGNAL. 1.81 3.4e-04  79
5   HALLMARK_TNFA_SIGNALING_VIA_N. 1.62 1.1e-03 188
```

On IFN-beta-stimulated data, IFN response must top the table — known answer, confirmed

A HABIT WORTH KEEPING

First step with any new tool:
run data whose answer you already know.

The ifnb answer is written in its experimental design. Only after it checks out do you trust the tool elsewhere.

04

Reading: NES, the plot, the leading edge

GSEA is not done until you can read all three

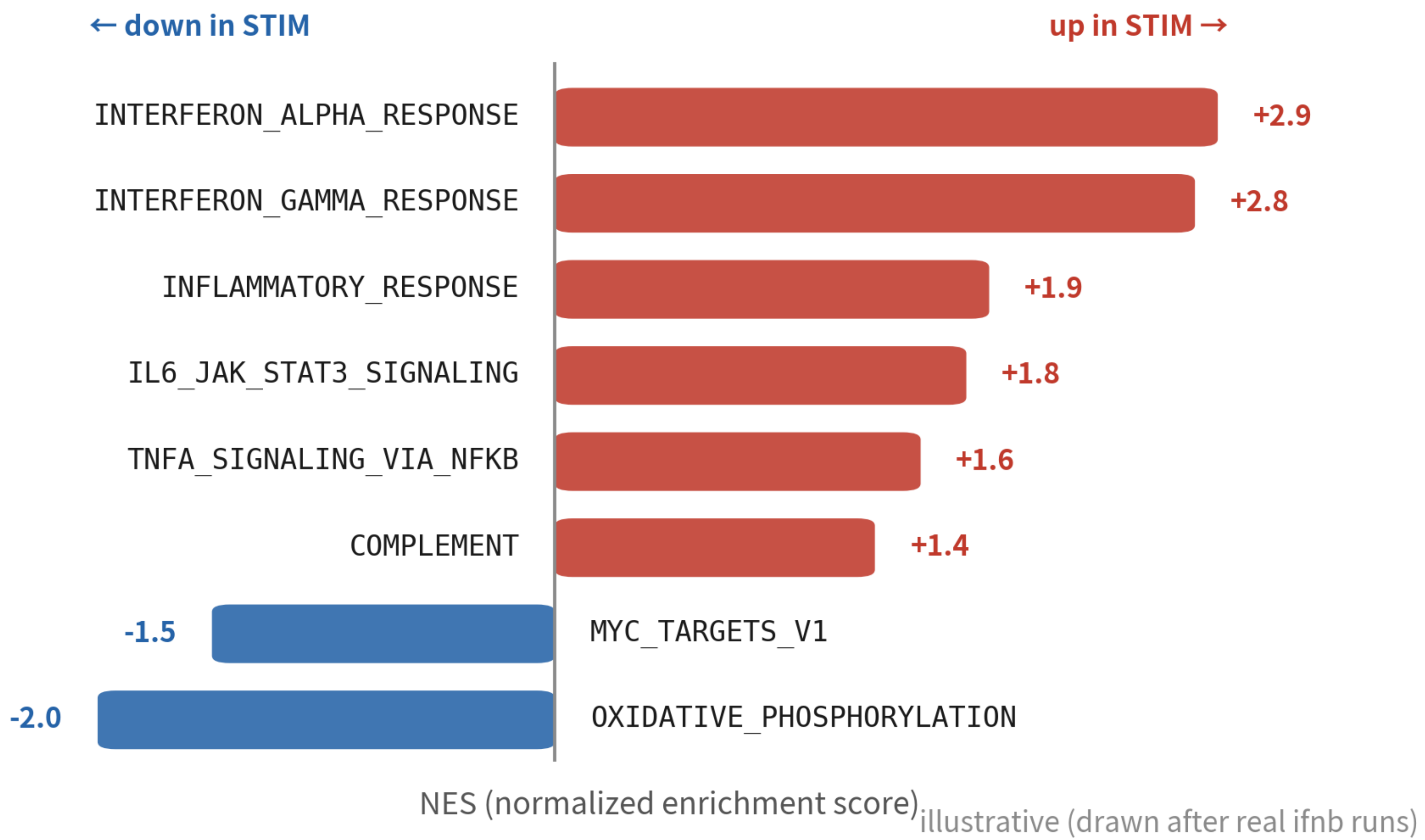
How to read NES

what NES gives you

sign = direction
(which end members crowd)

normalized for set size
→ comparable across sets

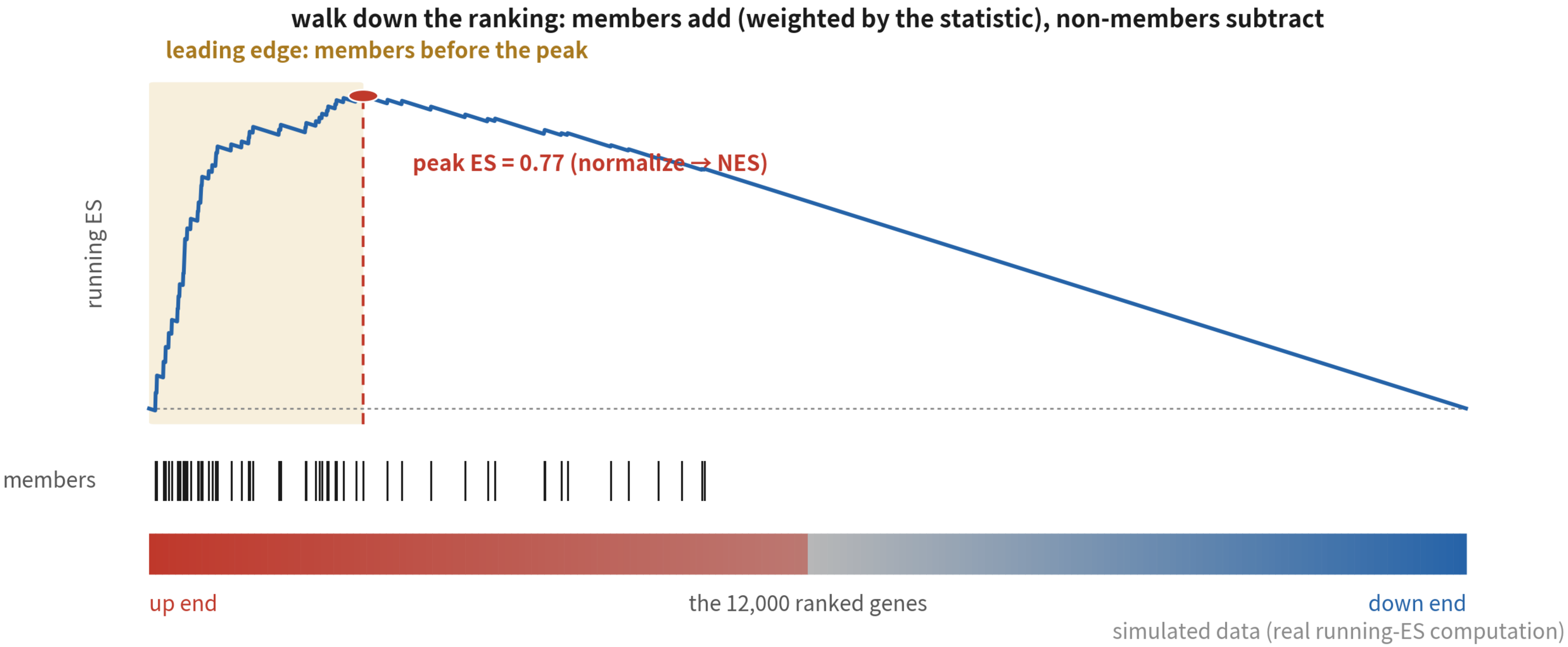
always read with padj



illustrative (drawn after real ifnb runs)

Direction, strength, cross-set comparability — and never without padj

The enrichment plot and the leading edge



Walk the ranking: members add, others subtract. Members before the peak = the leading edge

Plot it, then pull the leading edge

```
plotEnrichment(
 hallmark$HALLMARK_INTERFERON_ALPHA_RESPONSE, ranks) +
 labs(title = "HALLMARK_INTERFERON_ALPHA_RESPONSE")
ggsave("output/B13_gsea_ifna.png", width = 6, height = 4)

ia <- which(gsea$pathway ==
      "HALLMARK_INTERFERON_ALPHA_RESPONSE")
head(gsea$leadingEdge[[ia]], 8)
```

```
[1] "ISG15" "IFIT1" "IFIT3" "MX1"
[5] "OAS1" "RSAD2" "IFI6" "CXCL10"
```

leadingEdge is a list column: per pathway, the genes actually carrying the signal

Reporting one GSEA hit: the three-piece set

The numbers

NES direction and size, plus padj — never p alone

The plot

one enrichment plot, showing how members crowd

The leading edge

name the driver genes and reconnect to your biology

05

The ORA route: clusterProfiler and GO

The list is yours — the background must be too

Prepare the list and the background together

```
sig.up <- rownames(pb)[!is.na(pb$padj) &
                  pb$padj < 0.05 &
                  pb$log2FoldChange > 1]
length(sig.up)          # 上調且效應夠大的清單

universe <- rownames(pb)  # 背景 = 這次檢定過的基因
length(universe)
```

```
[1] 473
[1] 11143
```

The universe is not optional — it is half of ORA. Omit it and the tool substitutes the whole genome

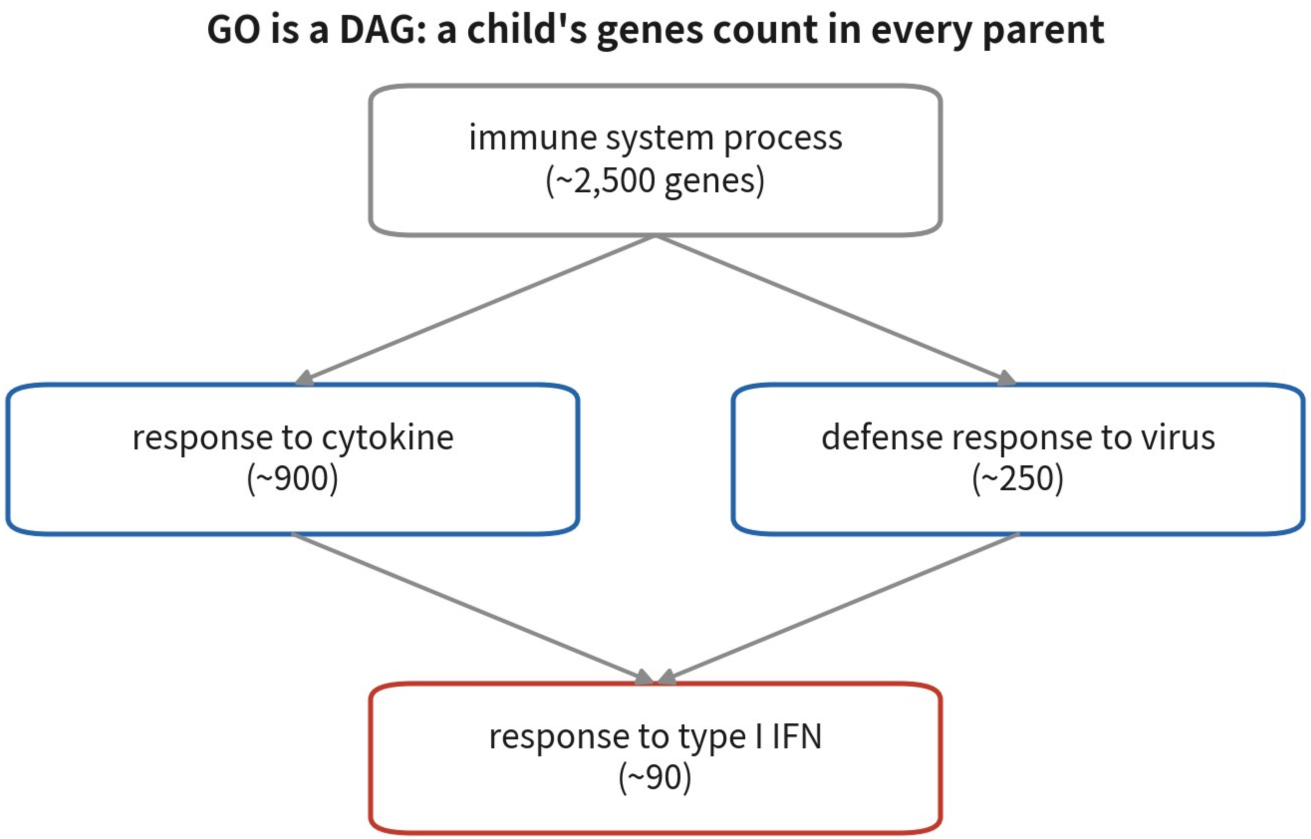
enrichGO and simplify

```
# BiocManager::install(c("clusterProfiler","org.Hs.eg.db"))
library(clusterProfiler)
library(org.Hs.eg.db)
ego <- enrichGO(gene     = sig.up,
        universe = universe,   # 關鍵參數
        OrgDb    = org.Hs.eg.db,
        keyType  = "SYMBOL",
        ont      = "BP")
ego <- simplify(ego)     # 合併語義高度重疊的條目
as.data.frame(ego)[1:3,
 c("Description","GeneRatio","p.adjust")]
```

```
                           Description GeneRatio p.adjust
GO:0051607   defense response to virus    58/442  1.2e-38
GO:0009615           response to virus    61/442  3.5e-37
GO:0034340 response to type I interferon  42/442  8.9e-33
```

GeneRatio's denominator, 442: list genes that map to GO annotations

GO is a hierarchy: parents share the child's genes



consequence: the same IFN genes

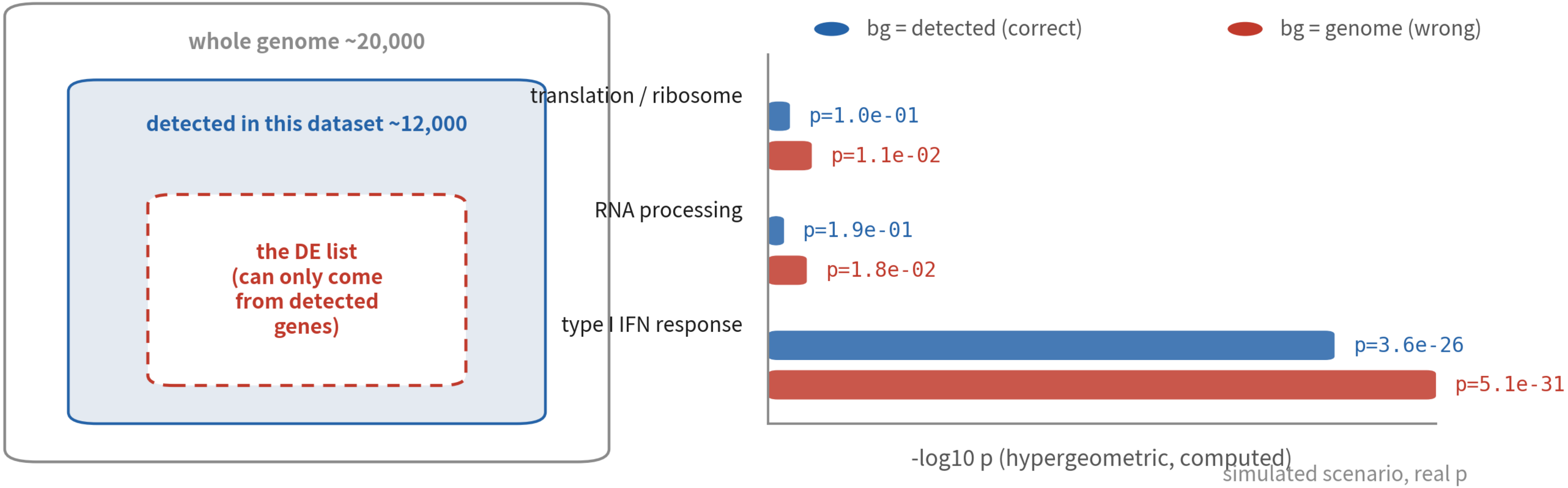
light up "IFN response", "defense response", "cytokine response", "immune process" at once

→ 40 rows that say one thing

fix: simplify() the terms and report by theme (see traps)

One batch of IFN genes lights a whole lineage — 40 rows, one story

The background decides the p value



swap the background and housekeeping pathways explode — the signal did not change, the denominator did

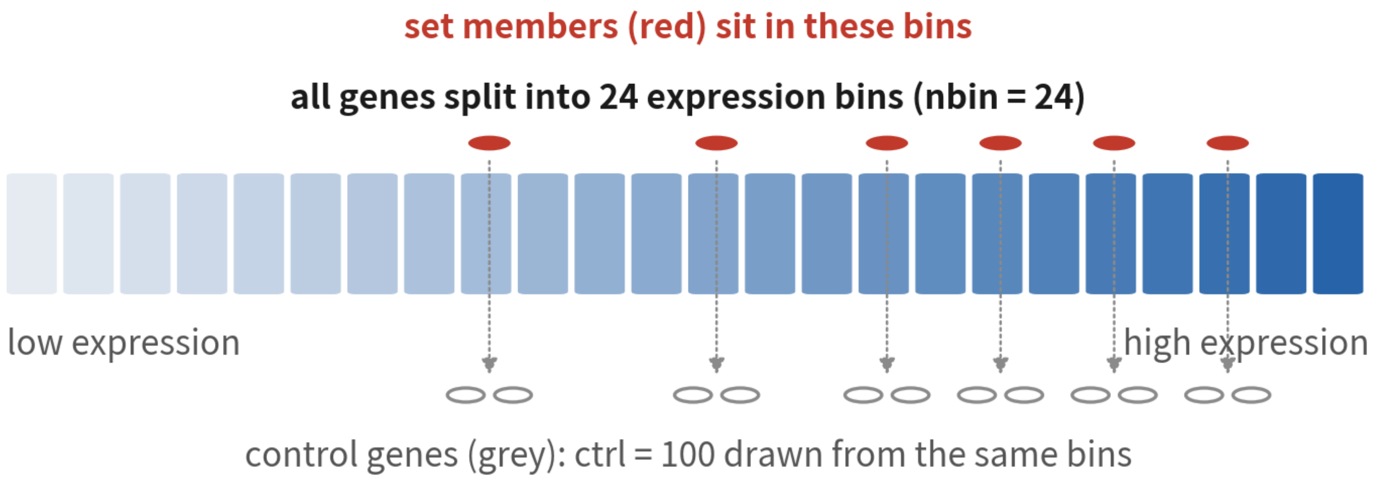
scRNA-seq detects ~12k genes — that is your background, not 20k

06

Cell level: paint scores back onto the UMAP

AddModuleScore and AUCell

How AddModuleScore works



each cell's score

mean(set expression)
− mean(control expression)

subtracting controls corrects depth and baseline; scores are relative — zero is just the reference

AddModuleScore: one gene set → one score per cell, stored in metadata

Mean of the set minus mean of bin-matched controls — the score is relative

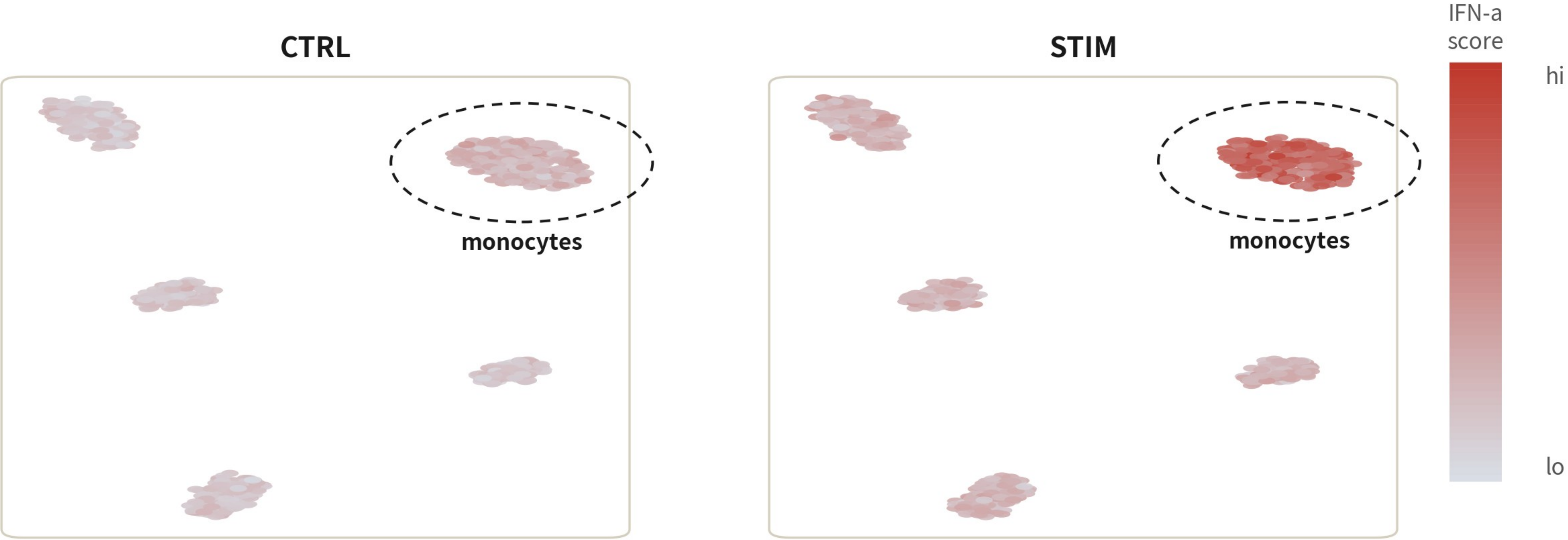
Score the cells, map the score

```
ifnb <- readRDS("output/ifnb_b11_integrated.rds")
ifnb <- JoinLayers(ifnb, assay = "RNA")

ifn.genes <- hallmark$HALLMARK_INTERFERON_ALPHA_RESPONSE
ifnb <- AddModuleScore(ifnb,
       features = list(ifn.genes),
       name = "IFNa_score")  # 預設 nbin = 24, ctrl = 100
FeaturePlot(ifnb, "IFNa_score1", split.by = "stim")
```

Column names get numbered: IFNa_score1. More sets in, more columns out

Two maps, one scale



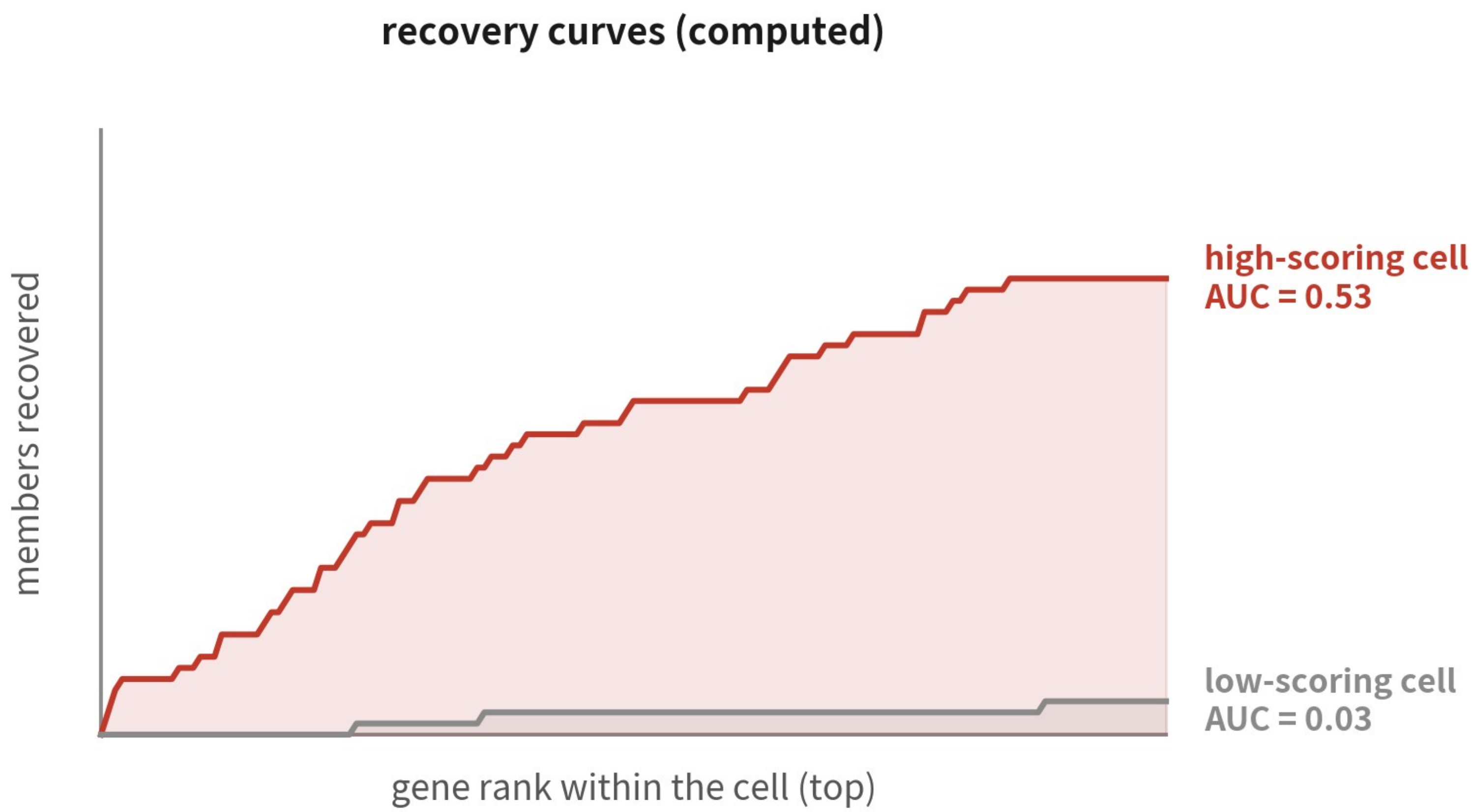
simulated data

same map, same scale: stimulated monocytes light up — the signal now has an address

Pseudobulk said CD14 Mono responds; this view shows where that lives on the map

AUCell: ranks instead of means

simulated data



1. rank genes within each cell (ranks only, not values)

2. how fast do set members appear in the top 5%?

3. area under curve = AUC → robust to depth / normalization

AUCell scores by ranking, not averaging — the engine inside SCENIC

Within-cell ranking, AUC over the top 5% — robust to depth and normalization

Three levels, three questions

Ranking level: GSEA

which programmes moved, overall, in this contrast

List level: ORA

which known sets my list resembles

Cell level: module scores

which cells on the map are running the programme

07

Where people get hurt

Three traps — each demonstrated for real

Trap 1: whole-genome background

What it is

Running enrichGO without a universe, so the whole genome annotation becomes the background.

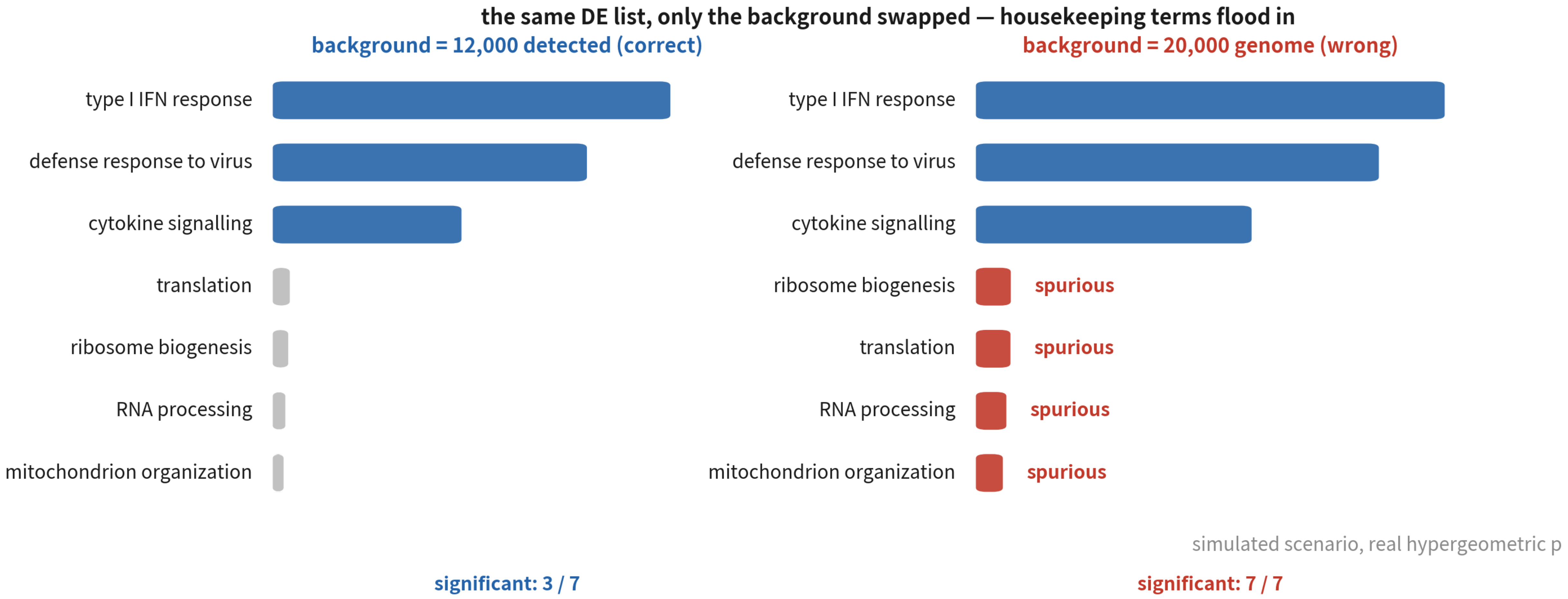
Why it happens

The argument is optional; it runs fine without it — and returns even more hits.

What it costs you

Everything scRNA-seq can detect turns 'significant' — translation, ribosome, metabolism — and drowns the real signal.

Live demo: one list, two backgrounds



The trap section of the R script reproduces this: both the count and the ranking change beyond recognition

Trap 2: enriching your own markers

What it is

Using a cluster's top markers as a gene set, then scoring or testing the same data with it.

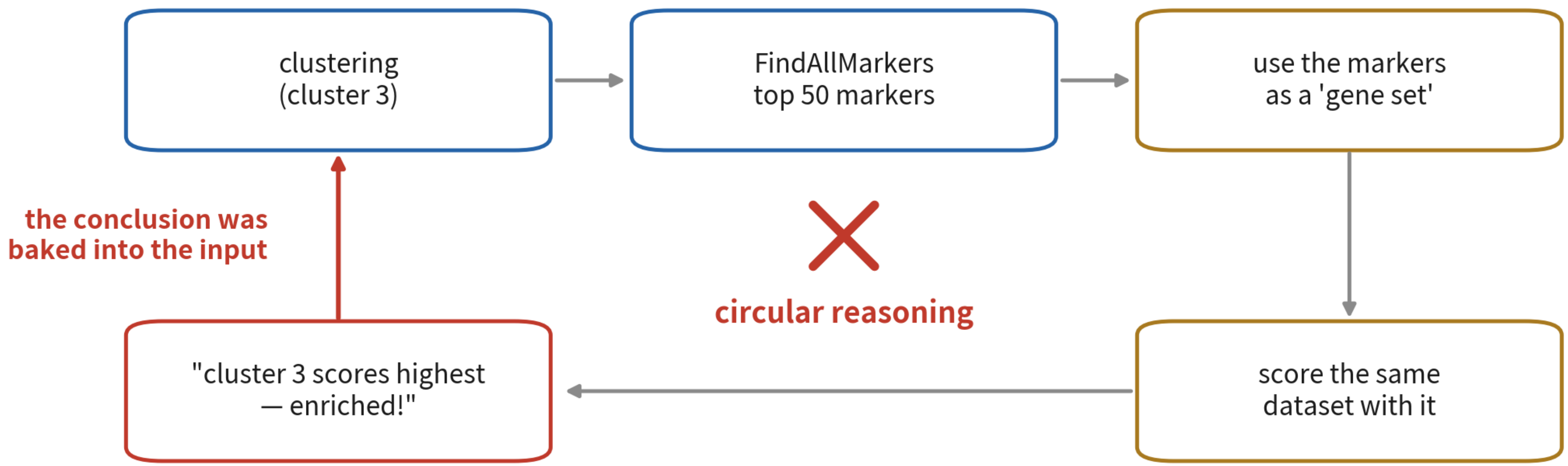
Why it happens

The pipeline runs and the plot looks great — of course that cluster scores highest.

What it costs you

Circular reasoning: the conclusion was baked into the input. Zero information, yet it gets published as 'validation'.

What the circle looks like



the right way

gene sets must come from knowledge independent of this dataset:

MSigDB / literature / signatures from other data

your own markers may name clusters — never validate them

Gene sets must come from knowledge independent of this dataset: MSigDB, literature, other data

Trap 3: reporting the top-20 by p

What it is

Sorting the GO table by p and pasting the top 20 into the paper.

Why it happens

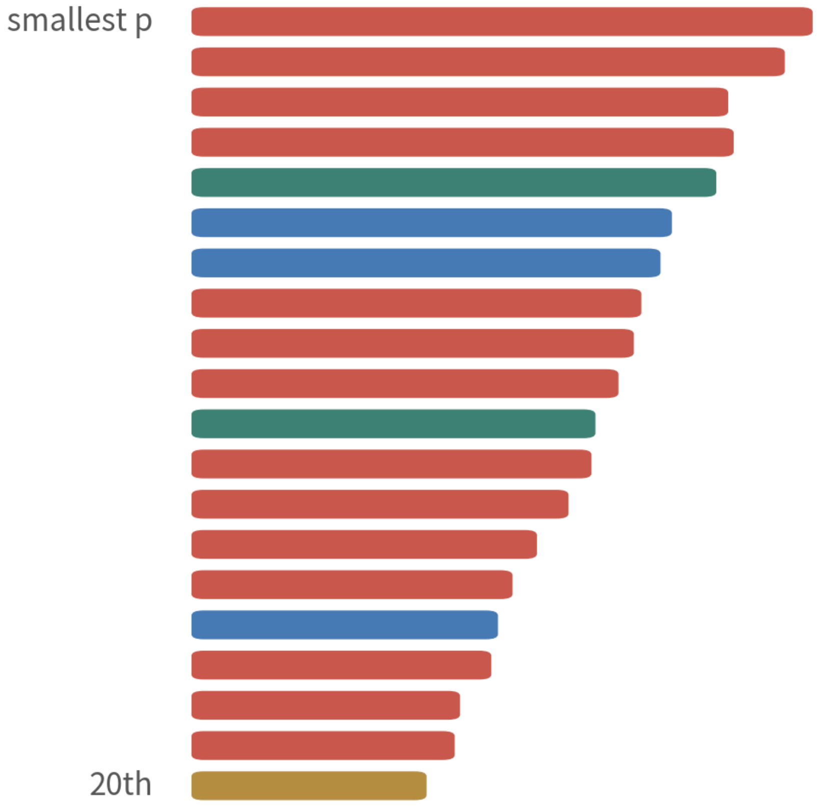
You have to truncate somewhere, and cutting by p feels objective.

What it costs you

Parent-child double counting: a dozen of the 20 restate one finding — and the door opens to cherry-picking whichever term fits the story.

Converge: report themes, not rows

GO table: top 20 by p value



red = variations on interferon / antiviral

collapse by theme

report themes, not rows

simulated data

1–2 representative terms per theme, with direction and leading edge

type I IFN / antiviral response

14 terms; NES all up

antigen presentation (MHC I)

3 terms

chemokines & leukocyte migration

3 terms

simplify() the terms, cluster rows into themes by NES direction and semantics, then pick representatives

TRAPS, SUMMED UP

What enrichment produces best is output that merely looks like a conclusion.

Right background, independent list, converged terms — only past all three gates does output become conclusion.

07+

Strategy Room: Complex Samples

The same fgsea, on a multi-patient tumour DE table — which gene sets, what to rule out first, how many runs

Tumour gene sets: beyond Hallmark, the meta-programs

Hallmark still opens; 'which state is this malignant cell in' takes sets curated by tumour research itself

The question	Which sets	Notes
Generic programmes: hypoxia, proliferation, EMT…	the 50 Hallmark sets (built into msigdbr)	the opener stays: scan once, set direction
GBM malignant-cell states	the four Neftel 2019 states: MES / AC / OPC / NPC-like	from the paper's supplement; after split, same format as Hallmark
State programmes of other cancers	melanoma Tirosh 2016 (MITF/AXL); head-and-neck Puram 2017 (p-EMT)	no copying across tumours — Neftel on liver cancer runs, and means nothing
Not tied to one cancer type	pan-cancer ITH meta-programs (curated by Gavish 2023)	the generic option for multi-cancer or undecided settings
Caution: state score ≠ malignancy evidence	these programmes are mostly defined within confirmed-malignant cells	a high score is a state tendency; malignancy calls rest on CNV (B16)

Dissociation stress and hypoxia: rule out, then read

Paying the debt the B5 strategy room left: only past these checks may either signal enter a conclusion

The dissociation signature

FOS/JUN/HSP is the enzymatic footprint; scored, it dots every cell type a little — not a subpopulation, not a pathway

Hypoxia: real and fake

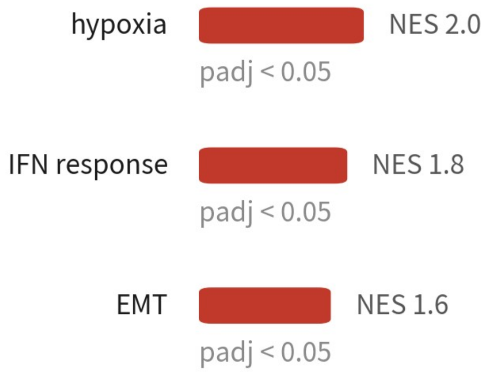
tumour cores genuinely lack oxygen (SLC2A1 up log2FC 5.7 in the GBM data) — but ex vivo ischemia mimics it

Two checks

does direction match the known answer (cores should be hypoxic); does it light every type like a microenvironment, or travel with the stress signature

Per-patient GSEA: only cross-patient consistency counts

four patients merged, one GSEA



all three look equally solid —
you cannot tell who carries what

rerun
per patient

the same test, once per patient

	pt 1	pt 2	pt 3	pt 4	
hypoxia	+2.1	+1.8	+2.4	+1.6	✓ all four agree — conclude
IFN response	+0.3	+2.9	-0.2	+0.4	⚠ one patient carries it — hypothesis only
EMT	+1.9	-1.7	+1.2	-2.0	✗ directions clash — do not report

red = positive NES in that patient (higher in core), blue = negative; numbers are NES

a conclusion must be able to say 'consistent in 4 of 4 patients' — the patient stays the unit, at the enrichment level too

schematic (simulated NES); scenario follows the GSE84465 GBM core-vs-margin data

A merged table cannot tell four-of-four from one-patient-driven — rerun per patient, lay the NES out, and it is obvious

ONE-LINE SUMMARY

GSEA asks direction, ORA asks the list, module scores ask where — the answers must agree.

All three levels of ifnb point to one sentence: stimulated monocytes switched on the type I IFN programme.

Check yourself 1

What input does each of ORA and GSEA take?

A Both take a list of significant genes

B ORA takes a list plus a background; GSEA takes a ranking of all genes

C ORA takes a ranking; GSEA takes a list

D Both take the expression matrix

Answer B. ORA is the urn model — does the list hold more members than chance? — so it needs the list and the background that defines chance. GSEA skips the cutoff and ranks every tested gene, asking where members crowd.

Check yourself 2

fgsea reports NES +2.9, padj 1e-23 for a pathway. The right reading is?

A Every gene in the pathway is significantly up

B Members crowd the up end, with enrichment normalized for set size

C The pathway is expressed 2.9-fold over background

D 2.9% of members are on the list

Answer B. NES describes the members' collective shift along the ranking: positive means crowding the up end, normalized for set size so sets are comparable. It never claims every member is significant — check the leading edge for that.

Check yourself 3

You ran enrichGO without a universe and got 200 hits full of translation and metabolism. First move?

A Pick the hypothesis-relevant terms for the paper

B Tighten the p cutoff to 0.01

C Set the universe to the genes actually tested and rerun

D Switch to KEGG and rerun

Answer C. A wall of housekeeping terms is the signature of a whole-genome background: undetectable genes flooded the denominator. A is cherry-picking, B just shortens a wrong list, D swaps databases while keeping the wrong background. Fix the background first.

NEXT EPISODE

B14: Cell-cell communication — ligands, receptors, CellChat

Enrichment tells you what a population is doing. But once monocytes fire up the IFN programme, what do they say to the T cells?

Next episode pairs ligands with receptors, infers a communication network — and spells out where that inference ends.